

New Pitch Detector

It is not uncommon when playing a music synthesizer for a struck note to fail to sound. This occurs primarily when VCFs and VCAs are driven by automatic envelopes (derived from the FORMANT ADSRs) and more than one key is pressed. The NPD helps to facilitate playing on the synthesizer keyboard by ensuring that the envelope(s) are triggered with every keystroke that results in a change in the KOV (Keyboard Output Voltage)! The "swallowing" of notes in the FORMANT (as well as in many other music synthesizers) can have several causes. Firstly, the ADSRs are subject to a certain degree of inertia; that is, the Gate signal must possess a certain minimum duration (a minimum Attack and Release time of approximately 20 ms) in order for any appreciable envelope to be generated at all. However, the culprit may also be "faulty" potentiometers within the ADSR units—specifically those with a residual resistance exceeding 10 Ω .

However, the issue usually simply stems from the fact that the player—whether intentionally or unintentionally—presses two or more keys simultaneously (in the case of the FORMANT keyboard, only the lowest-pitched note is audible). Naturally, no gate pulse can be triggered under these circumstances, as at least one key contact within the gate circuit remains closed at all times. Consequently, the VCFs and VCAs cannot "open up," since the envelope signals fail to materialize. The solution to this problem lies in the NPD (New Pitch Detector)—a device whose function can be loosely described as a "New-Key-Pressed Detector." Thanks to the NPD, it becomes possible to generate a "new" envelope with every "new" note, even when playing in a "legato" style (i.e., connected playing). The need to "hop" across the keys—a technique often detrimental to the precise reproduction of musical sequences—is now a thing of the past!

The block diagram in Figure 1 illustrates the basic structure of the New Pitch Detector. The circuit consists essentially of the "Pulse Generator" and the "Gate Follower."

Pulse Generator

To "detect" whether a new key has been played, the NPD differentiates the voltage changes of the KOV. However, for these changes to be reliably detected, the KOV must not (yet) be influenced by gliding portamento effects. The impedance converter IC1—which prevents the interface's Sample & Hold circuit from being loaded—is followed by a high-gain amplifier built around IC2. This amplifier shapes the voltage changes from the KOV into short, sharp pulses. The signals at the output of IC2 are negative for descending note sequences and positive for ascending note sequences. The negative pulses are passed on via D1. However, if the output of IC2 switches to a positive potential, the signal travels via R4 to an inverter built around IC3 and is then passed on via D2. Consequently, a negative signal is present at R7 in all cases; this signal is further amplified by T1 and T2. At the non-inverting input of IC5, this signal is combined with the output of the interface's Gate circuit. This ensures that no Gate signal is generated when the most recently pressed key is struck again.

Gate Follower

The NPD can optionally be housed either in the keyboard console or in the module cabinet; the specific implementation of the Gate Follower varies slightly depending on the chosen location.

When the New Pitch Detector is installed in the keyboard console, the Gate Follower consists essentially of a single comparator (see Figure 2). The "switching threshold" is fixed at approximately +2.3 V by the voltage divider R11/R12. At output G2—much like at the output of the interface's Gate circuit—a voltage of approximately +12 V to +14 V is present when a key is depressed; otherwise, the voltage remains at approximately +12 V to +14 V. The voltage waveforms at these two outputs diverge only during "legato" playing (see Figure 5); specifically, at output G2, the voltage momentarily (for approx. 2 ms) drops to a negative potential whenever a change in the KOV voltage occurs, thereby triggering a new envelope cycle within the contour generators. Switch S1 allows the user to select between the "normal" Gate signal and the NPD Gate signal.

If the New Pitch Detector is to be installed in the module cabinet, the circuit must be constructed according to the schematic shown in Figure 3. The only differences in this configuration are the addition of components R16 and D6. The diode clamps any negative output signals from IC4 to ground potential, while the voltage divider R15/R16 scales the positive signals down to approximately +5 V, thereby enabling direct control of the FORMANT ADSRs. Consequently, the signal at G2 corresponds—in terms of its voltage range—to the Gate signal generated by the interface receiver.

Circuit Construction and Testing

Figure 4 shows the printed circuit board and the component placement layout. Depending on where the NPD is to be installed, minor modifications may be necessary. When installing the New Pitch Detector inside the keyboard housing, the following connections must be established with the FORMANT Interface: Connect the junction point of T3, R4, and R5 to terminal K, and connect the output of the Gate circuit to input G. The dimensions of the PCB are such that the NPD circuit fits easily against the rear or side wall of a keyboard housing just 6 cm in height. If a new interface front panel (discussed elsewhere in this chapter) is not utilized—or if there is no space available for switch S1—the latter may be omitted. In this case, output G2 must be connected to the interface line within the "umbilical cord" cable assembly.

When installing the NPD inside the module housing, diode D5 and resistor R16 must be soldered into place. During assembly, the Gate output of the interface receiver is then connected to input G, and the KOV output to input K. However, this approach is only advisable if a Portamento switch (discussed elsewhere in this chapter) is also present, as reliable operation of the NPD cannot be guaranteed while Portamento is enabled. In this scenario, "legato" playing is possible only when Portamento is switched off. A 6-conductor connection between the keyboard housing and the module housing offers a solution to this issue. The sixth conductor is then

connected—at one end—to the KOV point (at the junction of T3, R4, and R5 on the FORMANT Interface) and—at the other end—to input K on the NPD.

For testing purposes, the circuit board must be fully populated and wired to the interface (or receiver). When a key is pressed, the voltage at output G2 should measure +12V to +14V (or +5V if installed in a manual housing); after the key is released, it should measure -12V to -14V (or 0V if installed in a manual housing). The proper functioning of the pulse generator can only be verified using an oscilloscope, as standard multimeters are too sluggish. It is also worth noting that the circuit should operate correctly without requiring any calibration. If the keys begin to "sing through" (sustain unintentionally) after the NPD has been installed, the key contacts will require readjustment.

It is difficult to fully appreciate the benefits of the New Pitch Detector based solely on a circuit description; the difference is best understood by experiencing it firsthand while playing.

References:

- M. Bertuch, Legato-Gate-ImpulsGenerator (LeGIG), Patchcord, No. 2, pp. 13–14
- Elektor, July/Aug. 1980, pp. 7–58/59

Parts list for picture 2.

Resistors (Carbon Film, 5%):

- R1, R9 = 8k2
- R2, R15 = 1 k
- R3 = 100 k
- R4, R6, R13 = 15 k
- R5, R10 = 39 k
- R7 = 4k7
- R8 = 27 k
- R11 = 12 k
- R12 = 2k2
- R14 = 1k8

Capacitors:

- C1 = 82 n (68 n + 15 n or 47 n + 33 n)
- C2 = 47 n
- C3, C4 = 470 n

Semiconductor:

- D1, D2, D3, D4 = DUS
- T1 = TUP

- T2 = DO
- IC1...1C4 = uA 741C

Miscellaneous:

- 1 miniature switch 1 x UM

The Bill of Materials for Figure 3 is identical to the Bill of Materials for Figure 2, with the following exceptions:

Resistors (Carbon Film, 5%):

- R14 = 3.3 k Ω
- R15 = 1.8 k Ω
- R16 = 1 k Ω

Semiconductors:

- D5 = DUG

Figures

- Figure 1. Schematic diagram of the New-Pitch-Detector. The KOV signal passes through an amplifier (A), an inverter (B), and a pulse generator (C) to the Gate Follower (D).
- Figure 2. Circuit diagram of the New-Pitch-Detector for installation in a keyboard housing. The interface signal is fed to input G. Via a switch S1, either the original gate signal (G = G1) or the New-Pitch signal G2 is available at the output.
- Figure 3. Circuit diagram of the New-Pitch-Detector for installation in a module housing. This circuit differs from the circuit shown in Figure 2 only by the addition of diode D5 and resistor R16.
- Figure 4. PCB layout and component placement diagram for the New Pitch Detector. Minor modifications are required depending on whether the PCB is installed in the module housing or the manual housing.
- Figure 5a. Voltage waveform at the output of the gate circuit of the FORMANT interface module, and the AD envelope waveform.
- Figure 5b. Voltage waveform at output G2 of the NPD circuit (when installed in the manual housing), and the AD envelope.